



ZCAMS Platform Governance

Super Admin & Company Admin Roles, Modules, and Support Tooling

Specification aligned to the ZCAMS Plotly Dash POC architecture: CFA onboarding, Bill of Lading capture, Reviewed BL, Z-SAD, GN 83 invoicing, CapitalPay check-out, payment settlement, cargo release, contracts, certificates, notifications, and support.

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1. Executive Summary

ZCAMS separates **platform governance** (Super Admin / ZAFFA) from **tenant administration** (Company Admin / CFA). Operational users (Declarant) execute clearance workflows but cannot change access policy. This document defines two dedicated dashboards, role assignment rules, permission boundaries, audit expectations, and five Admin support tools built from data already present in the ZCAMS SQLite schema.

2. Role Model & Assignment

2.1 Standard roles

Declarant = Agent. These are the same operational role in ZCAMS. The UI may show “Declarant / Agent”, but the stored role is *DECLARANT*. Legacy *AGENT* values are normalized automatically at login.

Role	Also known as	Scope	Assigned by	Primary purpose
SUPER_ADMIN	—	Entire platform (all CFAs)	ZAFFA IT / bootstrap only	Approve CFAs, monitor all transactions, configure policies, and
COMPANY_ADMIN	—	Single CFA company	Super Admin or existing Company Admin	Manage company users, documents, and limited operational
DECLARANT	Agent	Single CFA company	Company Admin (within Super Admin decision)	Operational clearance workflow only — no admin or platform

2.2 Role assignment rules

- Super Admin accounts are created outside self-service onboarding and cannot be assigned by Company Admins.
- On CFA onboarding (*/onboarding*), the first user is created as COMPANY_ADMIN with company status PENDING_APPROVAL.
- Super Admin must approve the company (*approve_company*) before full workflow access is granted.
- Company Admin may create Declarant (Agent) users and additional COMPANY_ADMIN users only if Super Admin has enabled that privilege.
- At least one active COMPANY_ADMIN must remain per company; suspension/deletion of the last admin is blocked.
- Super Admin may suspend any user, reset passwords, reassign roles, and override Company Admin decisions.

2.3 Permission matrix (target state)

Capability	Super Admin	Company Admin	Declarant / Agent
View own company dashboard	Yes (any company)	Yes	Yes
View platform-wide dashboard	Yes	No	No
Approve CFA onboarding	Yes	No	No
Manage company profile & certificates	Yes (all)	Yes (own)	Read only
Create / suspend company users	Yes (all)	Yes (own, limited)	No
Upload & review BLs	Yes (override)	Yes	Yes (operational only)

Capability	Super Admin	Company Admin	Declarant / Agent
Generate Z-SAD & GN 83 invoice	Yes (override)	Yes	Yes (operational only)
CapitalPay checkout & settlement view	Yes (all)	Yes (own)	Yes (own workflow only)
Issue cargo release	Yes (override)	Yes	Yes (operational only)
Contracts (create / sign oversight)	Yes (all)	Yes (own)	As configured by admin
View audit log	Full platform	Own company	No
View login / session history	Full platform	Own company users	Self only
Manage users or company documents	Yes (all)	Yes (own)	No
Access /admin or /super-admin modules	Yes	Yes (/admin only)	No
Change platform configuration	Yes	No	No
Support tickets — resolve escalations	Yes (all)	Own company	Create only

3. Super Admin Module (Platform Control Centre)

Route: `/super-admin` (dedicated shell; not shared with CFA operational nav). Super Admin receives full visibility across companies, users, workflow entities, and audit data stored in `zcams.db`.

3.1 Super Admin dashboard widgets

- Pending CFA approvals — companies with status `PENDING_APPROVAL` (from *companies*).
- Active CFAs & compliance score — approved companies with certificate completeness.
- Live workflow pipeline — counts for BLs, Reviewed BLs, Z-SADs, outstanding invoices, settled payments, releases.
- Today's activity — latest *audit_events* and *notifications* across all tenants.
- Open support escalations — *support_tickets* with priority High / status Open.
- Login activity — who signed in, when, from which IP (requires login audit enhancement).

3.2 Super Admin sub-modules

CFA Registry & Onboarding Queue — `/super-admin/companies`

List all companies; open onboarding packet (PACRA, TPIN, ZRA licence, certificates); Approve, reject, or suspend; trigger credential email to Company Admin.

Global User & Role Governance — `/super-admin/users`

Search users across tenants; assign `SUPER_ADMIN`, `COMPANY_ADMIN`, `DECLARANT` (Agent); suspend/activate; reset passwords; define permission templates per Company Admin.

Platform Audit Log — `/super-admin/audit`

Immutable view of *audit_events*: BL uploads, Z-SAD generation, invoice creation, payments, cargo release, profile edits, user CRUD, company approval. Filter by company, user, action_type, date. Export CSV.

Login & Session Monitor — `/super-admin/sessions`

New table recommended: *login_events* (user_id, timestamp, ip, user_agent, success). Super Admin sees failed attempts, concurrent sessions, and forced logout.

Transaction Monitor — `/super-admin/transactions`

Cross-tenant timeline: BL → Reviewed BL → Z-SAD → Invoice → Payment → Cargo Release. Drill-down to entity; Super Admin may correct status (with mandatory audit reason).

Invoices & CapitalPay Oversight — `/super-admin/invoices`

All invoices and checkout URLs; retry CapitalPay signing; mark settled manually in exception cases.

Contracts & Compliance — `/super-admin/contracts`

Cross-company contract register; signature status; certificate expiry alerts.

Notifications & Broadcast — `/super-admin/notifications`

Platform-wide feed; send maintenance or policy broadcasts to all CFAs.

Support Command Centre — `/super-admin/support`

All tickets; assign to ZAFFA agent; link to BL/invoice; close with resolution code.

Platform Settings — `/super-admin/settings`

GN 83 parameters, CapitalPay mode, OCR provider, email templates, feature flags, Company Admin permission caps.

4. Company Admin Module (Tenant Administration)

Route: **/admin** (Company Admin dashboard; distinct from operational */dashboard*). Company Admin manages access and company-side documents within limits set by Super Admin. They do not see other CFAs unless explicitly granted a cross-tenant support role.

4.1 Company Admin dashboard widgets

- Company compliance — profile completeness, certificates on file, banking details.
- Team overview — active/suspended users by role (from *users*).
- Workflow health (company-scoped) — pending BLs, invoices awaiting payment, releases pending.
- Recent company audit — last 20 *audit_events* for this *company_id*.
- Open support tickets raised by the company.

4.2 Company Admin sub-modules

Access Control — </admin/access>

Create Declarants; promote to Company Admin if allowed; suspend users; cannot create Super Admin; cannot exceed Super Admin permission template.

Company Profile & Banking — </admin/company>

Edit company details, logo, contact channels, bank account for invoice beneficiary fields.

Certificates & Documents — </admin/certificates>

Upload/replace PACRA, TPIN, ZRA licence, agency certificates; Super Admin may lock documents after verification.

Operational Oversight (read + limited edit) — </admin/operations>

Company-scoped BL, Reviewed BL, invoice list; correct OCR fields; re-upload BL when permitted; cannot delete settled financial records.

Contracts Administration — </admin/contracts>

Create contracts, resend OTP, track signature status for importers.

Notifications & Alerts — </admin/notifications>

Company notification feed; mark read; configure email preferences.

Support & Customer Service Tools — </admin/tools>

Five operational tools (Section 6) for call-centre support.

4.3 Limits imposed by Super Admin

- Maximum number of Company Admins and Declarants per CFA.
- Whether Company Admin may edit certificates after ZAFFA verification.
- Whether Company Admin may cancel/re-upload BLs after Z-SAD issuance.
- Whether Company Admin may manually mark invoices as settled (normally CapitalPay only).
- Whether Agentic Mode auto-send to clients is allowed without human approval.
- Module visibility (e.g. hide GN 83 schedule, disable chat model).

5. Audit, Login Tracking & Security Requirements

5.1 Current POC gaps to close

- Auth is client-side (dcc.Store in localStorage) — production needs server sessions or JWT.
- *audit_events.user_id* often defaults to demo user — must record acting user on every mutation.
- *ip_address* column exists but is not populated — capture on login and sensitive actions.
- No role checks on most callbacks — enforce RBAC middleware before page render and on every write.
- Download routes for invoices/documents are unauthenticated — protect with signed URLs.

5.2 Recommended audit actions (minimum)

Action type	Entity	Logged fields
USER_LOGIN / USER_LOGOUT	users	user_id, ip, timestamp, success/failure
COMPANY_APPROVED	companies	approver_id, company_id, previous/new status
BL_UPLOADED / BL_REVIEWED	Bills_of_lading	bl_id, bl_number, actor
ZSAD_GENERATED / ZSAD_DETACHED	ZSADs	z_sad_number, reviewed_bl_id
INVOICE_GENERATED / INVOICE_CLEARED	INVOICES	invoice_number, amounts, CapitalPay ref
CARGO_RELEASED	reviewed_bls	reviewed_bl_id, bl_number
USER_CREATED / USER_SUSPENDED	USERS	target user, role, actor
CERTIFICATE_UPLOADED	certificates	document name, company_id
ADMIN_OVERRIDE	any	reason text (mandatory for Super Admin corrections)

6. Company Admin Support Tooling (Customer Service)

These five tools use existing ZCAMS tables and integrations. Each is designed for a typical customer-service call: the Admin enters one identifier (BL number, invoice number, email, or company name) and receives an actionable answer within seconds.

Tool 1 — Shipment & Clearance Tracker

Typical complaint: Customer says: “Where is my cargo? Has customs cleared it?”

Data sources: Data sources: bills_of_lading, reviewed_bls, z_sads, invoices, payments, notifications.

- Single search by BL number, Z-SAD number, or invoice number.
- Visual pipeline: Uploaded → Reviewed → Z-SAD issued → Invoice sent → Payment settled → Cargo released.
- Shows timestamps, responsible user (from audit_events), and blockers (e.g. unpaid invoice).
- One-click copy of status summary for email/WhatsApp to client.

Tool 2 — Payment & CapitalPay Resolver

Typical complaint: Customer says: “I paid but the portal still shows outstanding” or “The checkout link expired.”

Data sources: Data sources: invoices, payments, CapitalPay URN/checkout_url, notifications.

- Lookup invoice by INV number or BL; show GN 83 breakdown (std min fee, admin fee, VAT, total).
- Display CapitalPay status, secure link, due date, and last settlement attempt.
- Flag mismatch between payable_amount and CapitalPay checkout total.
- Actions (if permitted): resend checkout link, refresh CapitalPay session, escalate to Super Admin.

Tool 3 — User Access & Login Diagnostic

Typical complaint: Customer says: “I cannot log in” or “My declarant account is locked.”

Data sources: Data sources: users, companies, audit_events, login_events (new).

- Search by email or username; show role, status (ACTIVE/SUSPENDED), company affiliation.
- Show last successful login, failed attempts, and whether company is still PENDING_APPROVAL.
- Actions: trigger password reset email, unsuspend (within policy), verify demo vs production account.

Tool 4 — Document & Certificate Completeness Check

Typical complaint: Customer says: “Onboarding is stuck” or “We cannot generate invoices — missing documents.”

Data sources: Data sources: companies, certificates, company profile fields, compliance_score logic.

- Checklist: PACRA, TPIN, ZRA licence, bank details, logo, signed contract status.
- Highlight missing or unverified certificates blocking approval or invoicing.
- Show upload date and who uploaded; link to /admin/certificates to fix.

Tool 5 — Invoice Dispute & GN 83 Fee Explainer

Typical complaint: Customer says: “This fee is wrong” or “Why is VAT different from what I expected?”

Data sources: Data sources: cargo_items (gn83_category), gn83.py rates, invoices, bills_of_lading.

- Recalculate GN 83 from stored cargo category, weight, and containers.
- Side-by-side: system calculation vs issued invoice; explain admin fee (20%) and VAT (16%).

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- Show Gazette Notice category mapping and minimum fee USD for the cargo type.
 - Export PDF fee breakdown for client dispute resolution (reuse pdf_service styling).

7. Implementation Roadmap (POC → Production)

Phase	Deliverables	Priority
Phase 1	RBAC middleware; login_events table; fix audit user_id; /super-admin shell	Critical
Phase 2	CFA approval UI; platform audit log; transaction monitor	High
Phase 3	/admin dashboard; access control UI; certificate admin	High
Phase 4	Five CS support tools on /admin/tools	Medium
Phase 5	Permission templates; server sessions; secured downloads	Medium

Summary: Super Admin owns the platform — approvals, global visibility, audit, and policy. Company Admin owns the CFA tenant — users, documents, and customer-service tools within Super Admin limits. Declarants (Agents) execute operational clearance workflows only and cannot access admin modules. Implementing these modules closes the current gap where all roles share the same operational dashboard without enforced permissions.

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